De La Salle Holy Cross College

NOVEMBER EXAMINATION

2024

Grade 10

PHYSICAL SCIENCES

Paper 2: Chemistry

Examiner: E. Hove
Moderator: A. Kamedza

Time: 2 hours
Marks: 130

NAME:_____

PLEASE READ THROUGH THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 20 pages and a Data Sheet of 2 pages. Please check that your question paper is complete.
2. Read the questions carefully.
3. Answer ALL the questions on the question paper and hand it in at the end of the examination. Use the data and formulae whenever necessary.
4. Show your working in all calculations. Units need not be included in the working of calculations, but appropriate units should be shown in the answer.
5. Answers must be expressed in decimal format, not left as proper fractions.
6. Where appropriate, express answers to TWO decimal places.
7. It is in your own interest to write legibly and to present your work neatly.
8. THREE blank pages10; 19 and 20 are included. If you run out of space for a question, use these pages. Clearly indicate the number of your answer should you use this extra space.

	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Total
Mark								
Question Total	**20**	**16**	**17**	**10**	**24**	**15**	**28**	**130**

QUESTION 1 MULTIPLE CHOICE

Answer the multiple-choice questions on the answer grid given below. Make a clear (X) in the box corresponding to the letter that you consider to be correct. Every question has only one correct answer.

A	B	C	D

Here the option C has been marked as an example.

1.1	A	B	C	D
1.2	A	B	C	D
1.3	A	B	C	D
1.4	A	B	C	D
1.5	A	B	C	D
1.6	A	B	C	D
1.7	A	B	C	D
1.8	A	B	C	D
1.9	A	B	C	D
1.10	A	B	C	D

1.1 Choose the correct definition for the term *compound*.

A A non-conducting material that does not carry any charge.

B A substance that allows an electrical current to pass through it.

C A pure substance made up of two or more different elements that are chemically joined together in a fixed ratio.

D A combination of two or more substances, where these substances are not bonded (or joined) to each other. No chemical reaction occurs between the substances.

1.2 Which ONE of the given formulae contains the greatest number of atoms?

A H_2O

B Na_2S

C $Al(NO_3)_3$

D NH_4

1.3 A measure of the amount of energy released when an electron is attached to a neutral atom or molecule (in the gaseous state) to form a negative ion is …

A ionisation energy.
B electronegativity.
C electron affinity.
D density.

1.4 Which group of the Periodic Table contains the element bromine?

A The alkali metals.
B The halogens.
C The noble gases.
D The transition metals.

1.5 Which ONE of the following equations shows the correct first ionisation of an element?

A $X + 245\ kJ.mol^{-1} \rightarrow X^+ + e^-$

B $X \rightarrow X^+ + e^- + 245\ kJ.mol^{-1}$

C $X + e^- + 245\ kJ.mol^{-1} \rightarrow X^+$

D $X^+ + 245\ kJ.mol^{-1} \rightarrow X + e^-$

1.6 Consider the elements sodium, magnesium, aluminium and silicon.

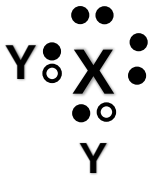
Which ONE of the given elements has the smallest atomic radius?

A Magnesium.
B Aluminium.
C Silicon.
D Sodium.

1.7 Which ONE of the following shows a molecule with polar covalent bonds?

A CS_2
B HBr
C O_3
D H_2

1.8 A chemical compound has the following Lewis diagram:



Which ONE of the following is the correct chemical formula of the compound formed when element **X** reacts with aluminium?

A $Al\mathbf{X}_2$
B $Al\mathbf{X}_3$
C $Al_2\mathbf{X}_3$
D $Al_3\mathbf{X}$

1.9 Which ONE of the following correctly relates the given substance to its lattice structure?

		O_2	**Fe**	**SiO_2**	**AgCl**
A		Molecular	Metallic	Giant covalent	Metallic
B		Molecular	Metallic	Molecular	Ionic
C		Molecular	Atomic	Molecular	Ionic
D		Molecular	Metallic	Giant covalent	Ionic

1.10 Consider the balanced chemical equation given below.

$3N_2 + 4Br_2 \rightarrow 2N_3Br_4$

If x moles of nitrogen are reacted, how many moles of the product is produced?

A $\frac{2}{3}x$

B $\frac{3}{2}x$

C 2x

D 4x

[20]

QUESTION 2

Silicon has the following three stable isotopes:

Isotope	Percentage abundance
^{28}Si	92,23
^{29}Si	4,67
^{30}Si	3,1

2.1 Define *isotop*e. (2)

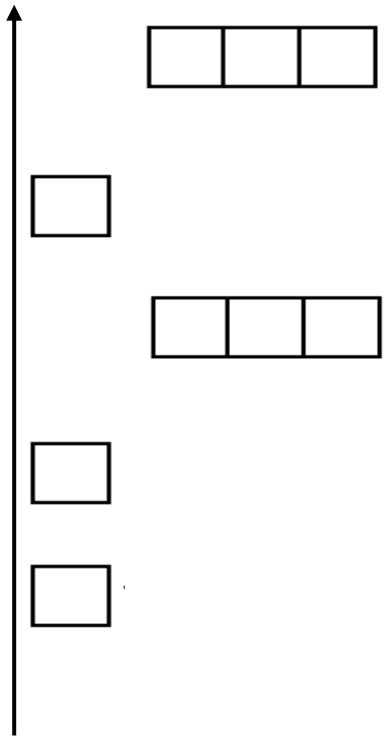
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2.2 Write down the atomic number for the element silicon. (1)

_____

2.3 Complete the following Aufbau diagram for the silicon atom. (3)



2.4 For the isotope ^{30}Si, write down…

2.4.1 the number of valence electrons. (1)

_____

2.4.2 the number of core electrons. (1)

_____

2.4.3 the number of neutrons. (1)

_____

2.5 Silicon ions in aqueous solutions can have a charge of +4 (Si^{4+}) or −4 (Si^{4-}), but the +4 ion is more stable in aqueous solutions. Write down the sp notation to give the electron configuration for the silicon ion with a charge of +4. (2)

_____

2.6 Calculate the relative atomic mass for the silicon. (2)

2.7 How does the fifth ionisation energy of silicon compare to the first ionisation energy of silicon? Write down GREATER THAN, EQUAL TO or LESS THAN. Explain your answer. (3)

_____

_____

_____

_____

[16]

QUESTION 3

Study the following list of substances.

chromium (VI) bromide	CO_2	NBr_3	CH_4
H_2O	Copper	sand in water	Toothpaste

3.1 Define *covalent bonding.* (2)

_____

_____

3.2 Write down the chemical name for NBr_3. (1)

_____

3.3 Identify from the given table a substance that:

3.3.1 conducts electricity well in solid form. (1)

_____

3.3.2 conducts electricity only in aqueous form. (1)

_____

3.3.3 exists as a liquid at room temperature. (1)

_____

3.3.4 is a heterogeneous mixture. (1)

_____

3.3.5 is an element. (1)

_____

3.4 Explain how particles are held together in the copper crystal lattice. (2)

_____

_____

_____

3.5 Draw a Lewis diagram for the CH_4 molecule. (2)

3.6 Write down a balanced chemical equation for the formation of chromium (VI) bromide from chromium and bromine. (3)

_____

3.7 Draw a Lewis diagram for chromium (VI) bromide. (2)

[17]

ADDITIONAL SPACE (ALL QUESTIONS)

REMEMBER TO INDICATE CLEARLY AT THE QUESTION THAT YOU USED THE ADDITIONAL SPACE TO ENSURE THAT ALL ANSWERS ARE MARKED.

QUESTION 4

Study the following equations and answer the questions that follow.

I : $Ni_2O_3 \rightarrow Ni + O_2$

II : $Cr_2O_7^{-2} + SO_3^{2-} \rightarrow Cr^{3+} + SO_4^{2-}$

III : $BaCl_2 + \mathbf{A} \rightarrow BaCO_3 + NaCl$

IV : $AgNO_3 + CaI_2 \rightarrow$

4.1 Define *oxidation*. (2)

_____

_____

4.2 Write down a balanced equation for reaction **I**. (1)

_____

4.3 For equation **I** write down:

4.3.1 the oxidation half reaction. (2)

_____

4.3.2 the oxidizing agent. (1)

_____

4.4 Write down the chemical formula of the spectator ions in equation **II**. (1)

_____

4.5 Write down the chemical formulae of the TWO products in equation **IV**. (2)

_____

4.6 Write down the chemical name of reactant **A** in equation **III**. (1)

_____

[10]

QUESTION 5

For a certain chemical reaction, you are given:

- Activation energy = 240 $kJ.mol^{-1}$
- Enthalpy of reaction (Heat of reaction) = – 45 $kJ.mol^{-1}$

5.1 Define *activation energy*. (2)

_____

_____

_____

5.2 Is the reaction given endothermic or exothermic? Explain your answer. (2)

_____

_____

_____

_____

5.3 Calculate the energy released during the formation of products. (2)

5.4 Draw an energy profile diagram for the information given. Include all given and calculated values. (5)

5.5 A catalyst is added to the reaction. Explain how the catalyst affects the rate of the reaction. (2)

_____

_____

_____

_____

5.6 Indicate the effect of a catalyst on your graph in QUESTION 5.4 on page 12. Label the graph catalysed graph. (2)

In a separate reaction, water is decomposed according to the balanced chemical equation given below.

$$2H_2O \rightarrow 2H_2 + O_2$$

Bond type	Bond energy ($kJ.mol^{-1}$)
O - H	464
H - H	436
O = O	496

5.7 Define *bond energy*. (2)

_____

_____

_____

5.8 Calculate the activation energy for the decomposition of water. (2)

5.9 Calculate the energy released during the formation of the products. (3)

5.10 Calculate the heat of the reaction. (2)

[24]

QUESTION 6

Ammonia (NH_3) is a weak base.

6.1 Define *a base*. (2)

_____

_____

Study the following equations and answer the questions that follow.

I : $H_2SO_4 + H_2O \rightarrow HSO_4^- + H_3O^+$

II : $HSO_4^- + OH^- \rightarrow H_2O +$ **P**

III : $HCl + NH_3 \rightarrow$ **Q** $+ NH_4^+$

IV : $NH_3 + H_2O \rightleftharpoons OH^- + NH_4^+$

6.2 Write down the chemical formulae of the TWO amphoteric substances in the given equations. (2)

_____

6.3 Identify the conjugate base in equation **I**. (1)

_____

6.4 Identify a conjugate acid in equation **IV**. (1)

_____

6.5 Write down the chemical formula of the substance represented by letter **P** in equation **II**. (1)

_____

6.6 Write down the chemical formula of the substance represented by letter **Q** in equation **III**. (1)

_____

6.7 Write down the chemical formula of a Bronsted Lowry acid in equation **III**. (1)

_____

6.8 Write down a balanced chemical equation for a complete chemical reaction between sodium hydrogen carbonate and nitric acid. (4)

_____

6.9 Describe a test that can be used to identify the gas produced in QUESTION 6.8. (2)

_____

_____

_____

_____

_____

[15]

QUESTION 7

Grade 10 pupils are provided with the following substances:

45 g of Glucose ($C_6H_{12}O_6$)	**2800 cm^3 of SO_2**
165,20 g of $Fe(MnO_4)_3$	**139,2 g of $MgSO_4.nH_2O$**

7.1 45 g of glucose ($C_6H_{12}O_6$) is dissolved in 240 cm^3 of water to make a sugar solution.

7.1.1 Define *concentration.* (2)

_____

_____

_____

7.1.2 Calculate the concentration of the sugar solution. (4)

7.2 Determine the number of SO_2 molecules in 2 800 cm^3 of SO_2 at STP. (4)

7.3 139,2 g of $MgSO_4.\mathbf{n}H_2O$ is strongly heated until a constant mass of 96 g is obtained. Determine the value of **n** in the given formula. (5)

7.4 165,20 g of $Fe(MnO_4)_3$ is decomposed in the presence of heat according to the following balanced chemical equation:

$$4Fe(MnO_4)_3(s) \rightarrow 2Fe_2O_3(s) + 12Mn(s) + 21O_2(g)$$

7.4.1 Give the chemical name for $Fe(MnO_4)_3$. (1)

_____

7.4.2 Calculate the mass of Fe_2O_3 produced in the given reaction. (6)

7.5 Iron (III) oxide reacts with Vanadium in a redox reaction to produce Fe and a compound with the following percentage composition:

56,044% vanadium and 43,956% oxygen.

7.5.1 Calculate the empirical formula for the compound produced during the reaction. (5)

7.5.2 Provide the chemical name of the compound calculated in QUESTION 7.5.1. (1)

_____

[28]

TOTAL MARKS: 130

ADDITIONAL SPACE (ALL QUESTIONS)

REMEMBER TO INDICATE CLEARLY AT THE QUESTION THAT YOU USED THE ADDITIONAL SPACE TO ENSURE THAT ALL ANSWERS ARE MARKED.